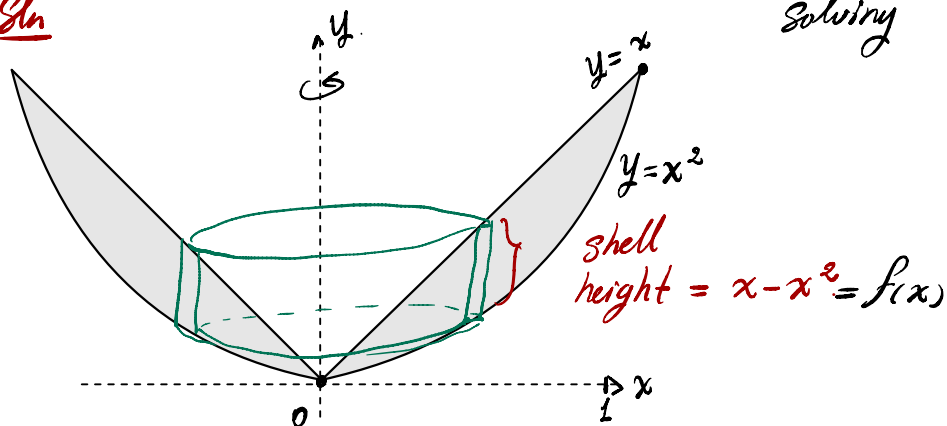


Example. Find the volume of the solid by rotating about the y -axis the region between $y=x$ and $y=x^2$.

Sln.



Solving $x^2 = x$

$$x(x-1) = 0 \Rightarrow x=0, x=1.$$

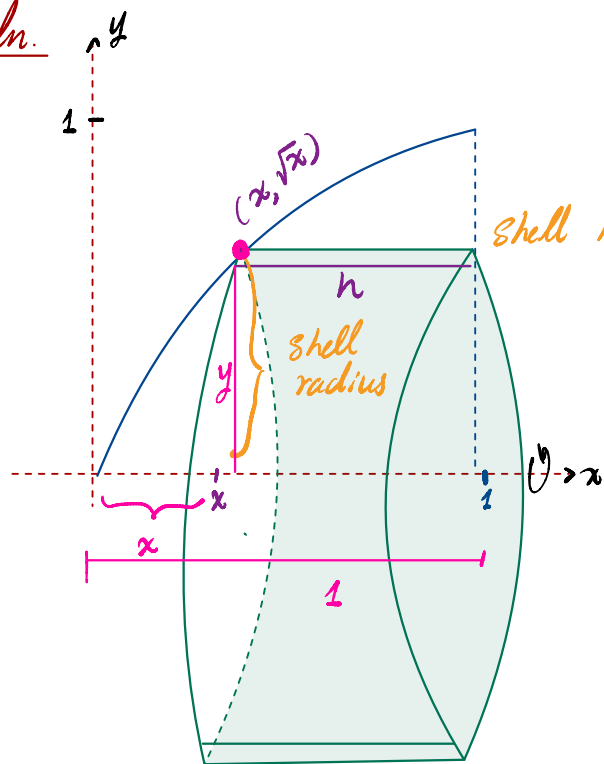
$$V = \int_a^b 2\pi x f(x) dx.$$

$$V = \int_0^1 2\pi x (x - x^2) dx = 2\pi \int_0^1 (x^2 - x^3) dx$$

$$V = 2\pi \left\{ \frac{x^3}{3} - \frac{x^4}{4} \right\}_0^1 = \frac{\pi}{6}.$$

Example Use the cylindrical shell to find the volume of the solid obtained by rotating about the x -axis the region under $y=\sqrt{x}$ from 0 to 1.

Sln.



shell height $= h = 1 - x$. Since $y = \sqrt{x}$ then

$$x = y^2$$

$$h = 1 - y^2.$$

$$r = y.$$

$$V = \int_0^1 2\pi y (1 - y^2) dy.$$

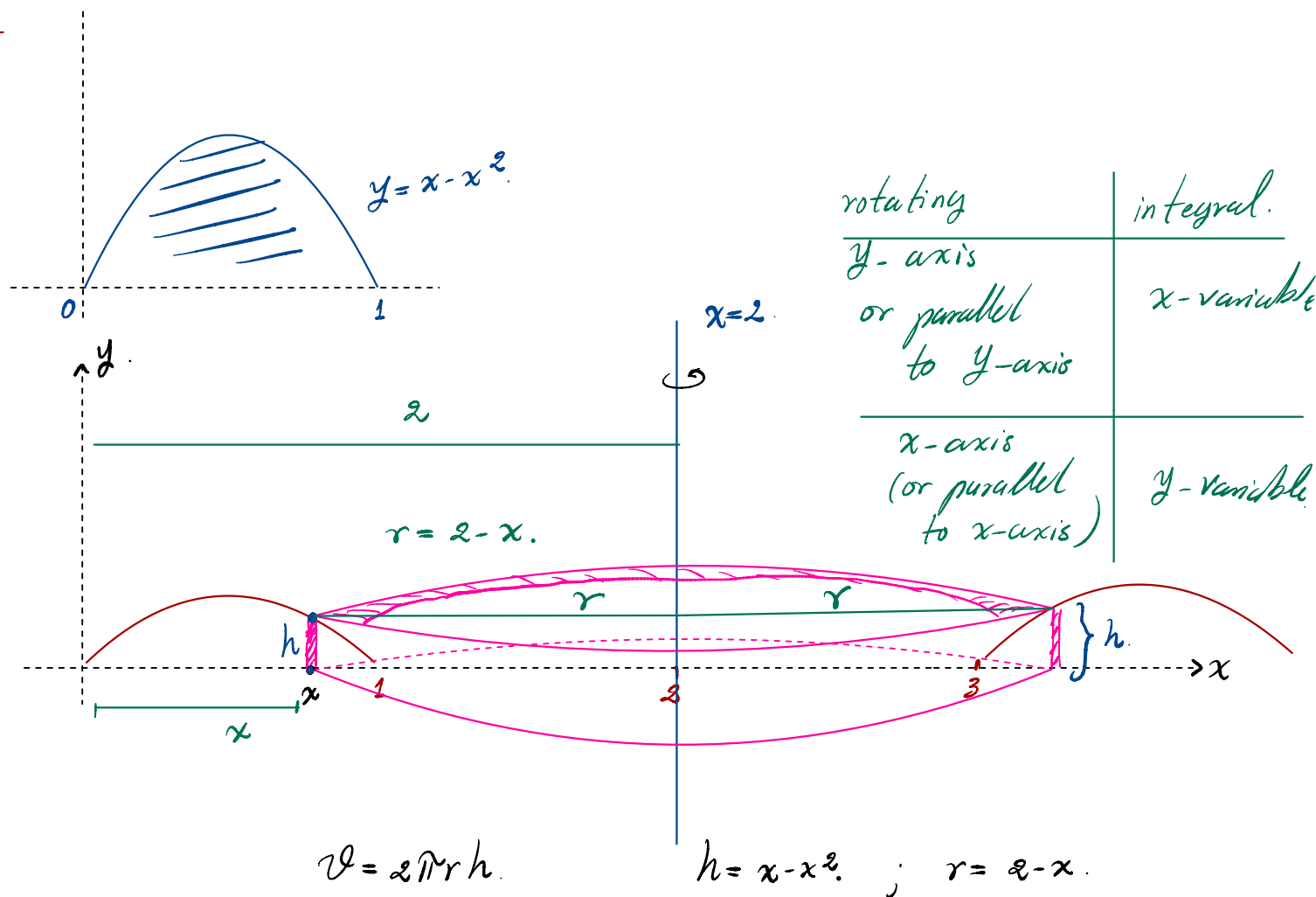
$$V = 2\pi \int_0^1 (y - y^3) dy.$$

$$V = 2\pi \left\{ \frac{y^2}{2} - \frac{y^4}{4} \right\}_0^1 = \frac{\pi}{2}.$$

Example. Find the volume of the solid obtained by rotating the region bounded by $y = x - x^2$ and $y = 0$ about the line $x = 2$.

$$x - x^2 = 0 \quad x(x-1) = 0 \Rightarrow x = 0, x = 1.$$

Sln.



$$\begin{aligned} V &= \int_0^1 2\pi(2-x)(x-x^2) dx = 2\pi \int_0^1 (2x - 2x^2 - x^2 + x^3) dx \\ &= 2\pi \int_0^1 (2x - 3x^2 + x^3) dx \\ &= 2\pi \left\{ x^2 - x^3 + \frac{x^4}{4} \right\} \Big|_0^1 \\ &= \frac{11}{2}. \end{aligned}$$

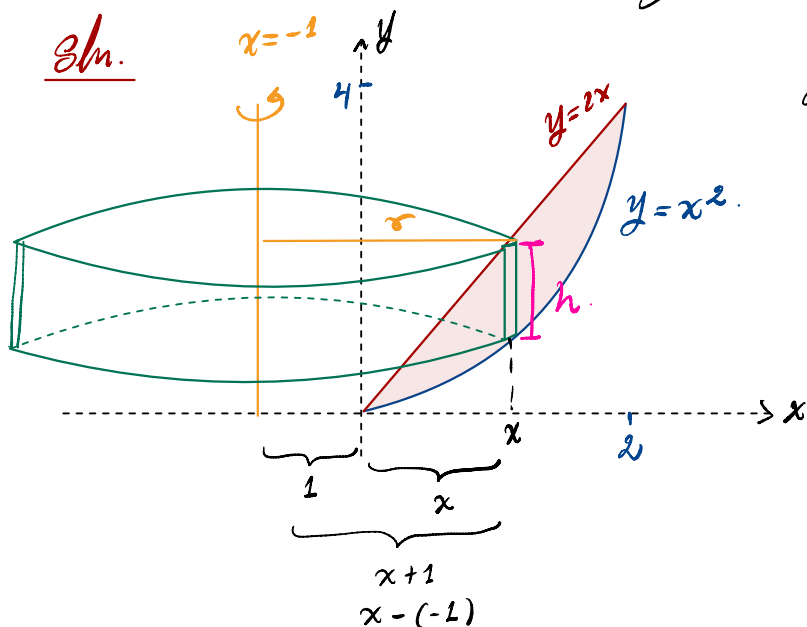
Example. The picture shows the region bounded by $y = x^2$, $y = 2x$. A solid is formed by rotating about $x = -1$.

Find the volume of the solid.

a). x as the variable of integration

b). y as the variable of integration.

Sln.

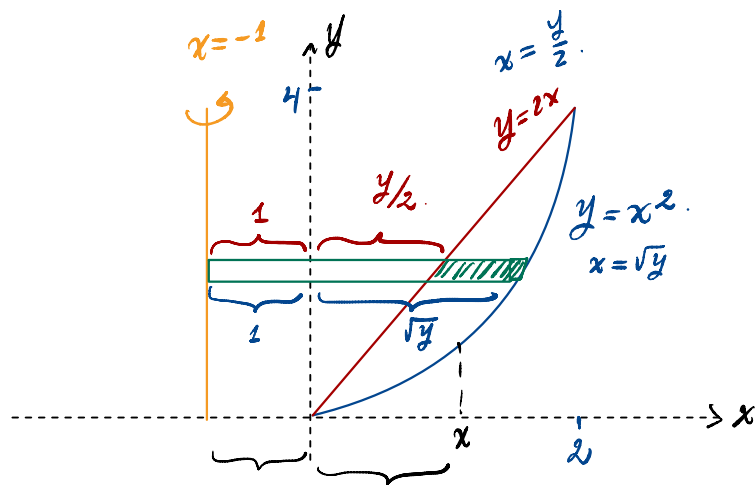


a). Using cylinders we have.

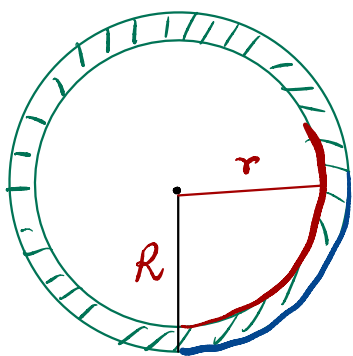
$$r = x+1. \quad h = 2x - x^2.$$

$$V = \int_0^2 2\pi(x+1)(2x-x^2)dx = \frac{16\pi}{3}.$$

b).



$$A(y) = ?$$



$$r = 1 + \frac{y}{2}$$

$$R = 1 + \sqrt{y}.$$

$$A(y) = \pi(1+\sqrt{y})^2 - \pi(1+\frac{1}{2}y)^2$$

$$= \pi((1+\sqrt{y})^2 - (1+\frac{1}{2}y)^2).$$

$$V = \int_0^4 \pi((1+\sqrt{y})^2 - (1+\frac{1}{2}y)^2) dy = \frac{16\pi}{3}$$